

PHYS503 Midterm Exam (April 13, 2025)

1. (20 points) Prove the gauge invariance of E&B fields under the transformation of

$$\vec{A}' = \vec{A} + \nabla\Lambda \text{ \& } \varphi' = \varphi - \frac{\partial\Lambda}{\partial t}.$$

2. (15 points) Prove the following relation. Use the Levi-Civita symbol in deriving the relation.

$$\nabla \times (\vec{f} \times \vec{r}) = 2\vec{f} + r \frac{\partial \vec{f}}{\partial r} - \vec{r}(\nabla \cdot \vec{f})$$

3. (15 points) From the Maxwell's equations & electromagnetic plane wave propagation in vacuum, show the following properties of the E&B fields

3.1. (5 points) E & B fields form the transverse wave

3.2. (5 points) E & B are perpendicular to each other

3.3. (5 points) [cB] is equal to [E] in magnitude

4. (50 points) Radiation from a center-fed linear antenna. Find the angular distribution of radiation power in terms of antenna length (d), current (I), and the wavelength ($1/k$).

Hint: you can use the angular distribution from an electric dipole moment:

$$\frac{dP}{d\Omega} = \frac{c^2 Z_0}{32\pi^2} k^4 |p|^2 \sin^2 \theta$$

Note that the current in the dipole radiator is expressed as

$$I(z)e^{-i\omega t} = I_0 \left(1 - \frac{2|z|}{d}\right) e^{-i\omega t}$$

